

# Neha NamiReddy

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## WORK EXPERIENCE

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### FedEx | Software Engineer - IT Intern

Jun 2025 - Aug 2025

- Maintained and enhanced a Python-based backend application using FastAPI, processing 1,000+ records from Excel and JSON with REST APIs and SQL, while resolving data validation, performance, and business-logic issues.
- Monitored performance, traffic, errors, and user activity with Datadog, establishing operational discipline through AWS EC2 instance management, CI/CD workflows, and recovery protocols, cutting deployment failures 40% and improving platform stability.
- Built real-time data processing pipelines that ingested atmospheric-style observation records as they arrived, accounting for late or missing data and flagging anomalies for immediate attention.
- Developed Power BI dashboards and a centralized Power Apps application backed by Azure SQL for 50+ internal teams, replacing fragmented manual reporting with unified visualization tools that streamlined data collection, reporting workflows, and decision-making efficiency by 50%.
- Built tooling and monitoring infrastructure to detect pipeline failures, data quality issues, and model inference slowdowns, ensuring visibility into operational health and enabling rapid incident response.

### University of Houston | AI/ML - Instructional Assistant

Jan 2025 - Present

- Built an AI-assisted grading validation pipeline using Python, Pandas, NLP embeddings, and cosine similarity to automate Excel-based assignment review for 250+ students, reducing manual grading checks by 75% and identifying missing, duplicate, and semantically similar submissions.
- Supported 25+ student teams through semester-long machine learning and deep learning projects, providing technical guidance on model selection, data preprocessing, and experiment design, improving project completion rates by 30%.

### Experian | Software Engineer

Dec 2023 - Jul 2024

- Optimized Python-based data workflows for processing, validating, and transforming data, while conducting code reviews, resolving Git merge conflicts, and enhancing CI/CD pipelines, increasing deployment efficiency by 20%.
- Optimized and tuned complex SQL queries by analyzing execution plans, indexing strategies, joins, and stored procedures, reducing average query response time by 63% and improving application performance for data-intensive workflows.
- Resolved recurring failures in AWS-based Apache Airflow deployment pipelines, reducing pipeline failure rate by 40% and improving DAG success rate to 90%, decreasing manual intervention incidents and strengthening system recovery.
- Led a LocalStack proof of concept (PoC) for local AWS cloud emulation, evaluating its adoption for offline development, testing, and CI/CD workflows.
- Leveraged Generative AI tools (Microsoft Copilot and Claude) to automate and accelerate technical documentation, requirements summarization, and reporting workflows, reducing preparation time by 60% while improving consistency and engineering productivity.

### Hyderabad Rock Sand | Software Engineer

Nov 2021 - Nov 2023

- Developed and maintained an internal enterprise application using Java, Spring Boot, REST APIs, and MVC, processing 10,000+ records across business workflows while performing performance optimization and memory analysis using stack traces and heap dumps, reducing peak heap usage by 25%.
- Enhanced React frontend components integrated with Spring Boot REST APIs, optimizing data-driven workflows and improving frontend performance, with a focus on clean UI design for operational dashboards.
- Increased JUnit/Mockito test coverage from 80% to 95%+ and leveraged ChatGPT for AI-assisted code review, debugging, and edge-case analysis, contributing to a 15% reduction in production bugs while strengthening application security with Checkmarx.
- Developed interactive Tableau dashboards to visualize sales trends, KPIs, and business performance, enabling real-time monitoring and data-driven decision-making for business stakeholders.
- Built and deployed an XGBoost supervised learning model in Python for sales prediction, applying feature engineering, model training, and validation on historical sales data to generate predictive insights and support data-driven forecasting, contributing to a 30% increase in sales.

## EDUCATION

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### University of Houston

Master of Science, *Computer Science*

Aug 2024 - May 2026

GPA: 3.93/4.0

## PROJECTS

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### Gene–Disease Hypothesis Generator

- Built an LLM-powered Retrieval-Augmented Generation (RAG) pipeline that retrieves relevant PubMed literature using sentence embeddings and FAISS, then uses BioGPT to generate evidence-grounded gene-disease hypotheses.
- Developed a biomedical NLP pipeline with NER, entity normalization, semantic retrieval, and evidence relevance scoring across 400+ literature passages. and built a Streamlit interface for hypothesis generation, evidence visualization, search, and expert feedback collection.

### Smart Yoga Trainer

- Built a real-time web application using computer vision for yoga pose classification that integrated on-site machine learning algorithms.
- Overcame challenges integrating pose detection into live video feeds by fine-tuning MediaPipe and OpenCV parameters, boosting processing speed by 30% and improving accuracy.

## SKILLS

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**Languages:** Java, Python, C/C++, SQL, JavaScript

**Frontend & Backend Development:** React, Node.js, Express, FastAPI, Flask, REST APIs, HTML, CSS, Bootstrap, MongoDB, MySQL, Spring Boot, UI Design, Visualization Tools

**Cloud, DevOps & Systems:** AWS, Docker, Kubernetes, Jenkins, CI/CD, Microservices, Linux, LocalStack, Airflow, Datadog, Distributed Systems, Data Pipelines, System Monitoring, System Reliability, System Recovery, Failure Detection

**Data & AI/ML:** Machine Learning, Deep Learning, NLP, LLMs, OpenCV, Pandas, NumPy, Spark, ETL, Power BI, Tableau, PyTorch, GPU Computing, ML Inference

**Tools & Practices:** Git, GitHub, Bitbucket, JIRA, Confluence, Agile, Postman, Maven, Excel, Google Colab

## CERTIFICATIONS & ACHIEVEMENTS

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- Amazon Web Services Cloud Practitioner - AWS
- Python for Data Science - NPTEL